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Interpretable multiscale algorithm unrolling network using learnable Gabor filters for fault diagnosis

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ABSTRACT

Intelligent fault diagnosis has emerged as a pivotal research area in intelligent maintenance systems, stressing interpretable intelligence and trustworthy decision-making for reliable and safe engineering applications. To this end, this study proposes the Interpretable Multiscale AlGorithm unrolling Network using learnable Gabor filters (IMAG-Net). Specifically, IMAG-Net establishes a physically optimization-unrolling framework that integrates learnable Gabor filtering with multiscale convolutional sparse coding, embedding the sparsity prior of fault-induced into a trainable neural architecture for adaptive and interpretable fault diagnosis. Distinct from traditional deep learning approaches, the proposed framework enhances interpretability and decision transparency through structured convolutional sparse coding and adaptive multiscale decomposition, where Gabor filters with learnable center frequencies automatically capture dominant signal components that correspond to fault characteristic frequencies and their harmonics. To further ensure physical fidelity, the learned multiscale representations are analyzed alongside sparsity-guided signal reconstructions derived from salient sparse activations, thereby validating the reliability and internal consistency of the network's decision logic. Experimental results show that the proposed method achieves a diagnostic accuracy exceeding 94% under high-noise conditions ($\sigma_n = 1.0$), consistently outperforming conventional deep learning baselines while preserving physically meaningful and interpretable diagnostic representations.

1. Introduction

Modern electromechanical systems have been extensively implemented in industrial applications, where bearings serve as critical components whose operational integrity directly governs system stability and lifespan of the system [1,2]. Research on bearing fault detection and health management is crucial to improve the safety, reliability and maintainability of this equipment [3,4]. Accurate identification of bearing faults not only increases operational efficiency, but also significantly minimizes unplanned downtime and financial losses. To achieve this, effective diagnostic techniques are essential, with vibration analysis emerging as a primary method due to its sensitivity to fault-induced changes.

Vibration signals serve as one of the most critical diagnostic features. Defects such as bearing wear and cracks alter vibration patterns, leading to changes in specific frequency components [5]. By analyzing vibration signals, the operational status of bearings

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can be effectively identified [6]. The time domain, frequency domain, and time frequency characteristics of vibration signals are of great significance for diagnosing different types of faults [7]. To further refine feature representation, advanced signal processing techniques have been continually developed. For instance, Jiang et al. proposed an adaptive composite weighted index, denoted as ACWIgram, to identify optimal demodulation bands by integrating impulsiveness and cyclostationarity [8]. Complementing statistical indices, sparse representation methods such as the multi-level refinement convolutional dictionary learning by Jin et al. have demonstrated strong capabilities in extracting weak fault signatures via frequency-domain sparsity [9]. However, these analytical methods often require manual expertise for hyperparameter selection, and achieving efficient end-to-end inference for massive datasets remains a challenge. In recent years, deep learning based intelligent fault diagnosis methods have leveraged the data characteristics of vibration signals to achieve high-precision automated diagnosis [10–12].

Intelligent algorithms represented by Convolutional Neural Networks (CNN) and Transformer architectures have made significant progress in the field of fault diagnosis, while the inherent 'black-box' presents the new challenge [13–15]. Unlike traditional analytical methods based on physical principles, deep neural networks automatically extract features and establish complex mappings from input data to output results [16,17]. The insufficiency of explicit mathematical formulations and clear decision-making processes in these models poses challenges for understanding how their predictions are derived. Consequently, the lack of interpretability hinders their practical deployment in safety-critical fields [18,19]. Additionally, most existing models primarily focus on learning features directly from raw vibration signals in the time domain. While time-domain analysis captures temporal patterns, it often fails to fully exploit the rich spectral characteristics inherent in vibration signals. These models often learn unknown frequency components within the network rather than establishing a clear correspondence with the characteristic frequencies of vibration signal faults.

Integrating signal processing priors into neural network architectures has emerged as an effective strategy for improving model interpretability and trustworthiness. Time–frequency representations, in particular, play a critical role in practical applications such as speech recognition [20,21]. Xu et al. conducted an in-depth study on the relationship between the learning dynamics of CNNs and frequency components, and proposed the Frequency Principle (F-Principle). Specifically, they observed that CNNs tend to fit target functions from low to high frequency components during training [22,23]. Regarding generalization, Wang et al. found that it is often influenced by the network's ability to capture unseen high-frequency components [24]. Building upon these theoretical insights, researchers have explored embedding learnable filters into network architectures. Mila et al. proposed the SincNet, a model that directly learns low-level speech representations from waveforms, enabling more effective capture of key frequency features in speech [25]. Advancing this paradigm, Liu et al. designed an interpretable FIRNet with learnable center frequencies to explicitly mimic the fault modulation principle [26]. Similarly, the Gabor filter, known for its selective ability to capture specific frequency bands, has been widely applied in various domains [27,28]. Parcollet et al. integrated complex Gabor filters with deep neural networks, replacing traditional CNN weight kernels to fully leverage the optimal performance of Gabor filters in time-frequency resolution [29]. Niknazar et al. designed an interpretable deep learning system based on Gabor filters and LSTM networks to classify EEG time series, achieving interpretability from microstructure to macrostructure [30].

Time-frequency methods have shown exceptional performance in bearing fault diagnosis. Liu et al. proposed a multiplicative convolutional residual network that leverages the intrinsic characteristics of time-frequency representations of vibration signals [31]. Chen et al. utilized time-frequency convolutions to process raw signals, achieving promising results [32]. Song et al. established multi-channel single-step decomposition induced by detected center frequencies based on multivariate variational mode decomposition and manifold learning, achieving fault feature enhancement [33]. To further address robustness challenges, Jia et al. proposed a multi-scale wavelet network, denoted as MWKG-Net, guided by spectral kurtosis to adaptively enhance impulse components [34], while another study by Jia et al. introduced a graph wavelet network to handle complex dynamics under time-varying speeds [35]. Nevertheless, a major limitation of these approaches lies in treating time–frequency processing merely as a preprocessing step, which often results in reduced interpretability in the subsequent network layers.

In addition, constructing trustworthy networks and employing post hoc interpretability methods are also critical strategies for achieving interpretability. ML-ISTA and AMP-Net have demonstrated superior performance in compressed sensing [36,37]. To extend this paradigm to fault diagnosis, An et al. introduced NISTA-Net, which integrates sparse priors into algorithm unrolling networks to enhance fault discrimination [38]. Nico et al. applied Shapley additive explanations (SHAP) as a post hoc interpretability method in the fault diagnosis domain, keeping the input domain in its original time-domain form while allowing the explanation domain to be selected in either the frequency or time-frequency domain, thereby effectively distinguishing fault samples [39]. Yan et al. combined Alternating Direction Method of Multipliers (ADMM) with the short-time Fourier algorithm, utilizing an interpretable weight matrix to visualize how weights evolve over time, validating the cyclostationarity of repetitive transients in fault vibration signals [40]. Notwithstanding, since most existing unrolling-based models are primarily tailored for compressed sensing, they may not fully capture the multiscale temporal structures inherent in complex vibration signals.

Furthermore, preserving physically meaningful interpretability under high-noise conditions remains a significant objective for data-driven methods. Conventional deep neural networks often encounter difficulties in explicitly capturing the bearing-specific fault characteristic frequencies inherent in vibration signals. This limitation frequently leads to degraded diagnostic performance when useful fault features are submerged by strong background noise or spectral aliasing. To address these combined challenges, which include the need for domain specificity in unrolling models and the noise robustness of general-purpose architectures, this paper proposes the Interpretable Multiscale AlGorithm unrolling Network using learnable Gabor filters (IMAG-Net).

The main contributions and highlights of this study are described as follows:

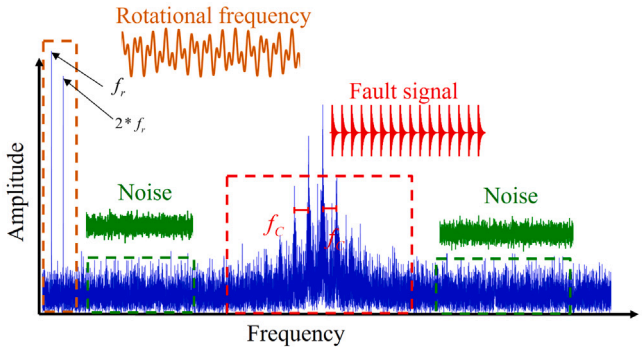


Fig. 1. Signal characteristics of the faulty bearing.

(1) An adaptive multiscale feature extraction mechanism is introduced, combining learnable Gabor filters with convolutional sparse coding to dynamically capture discriminative spectral and temporal components from vibration signals. The filters' center frequencies are optimized end-to-end, allowing the model to focus on fault-relevant bands. Experimental results show that replacing these learnable filters with fixed causes a decline in classification accuracy, thereby demonstrating the effectiveness of adaptive frequency feature learning.
(2) We construct a deep network by unrolling the iterative steps of a multiscale convolutional sparse coding algorithm. The integration of sparsity priors into the network dynamics constrains the model to focus specifically on fault-related time frequency features. Such a structured representation facilitates the identification of characteristic spectral components associated with bearing defects, thereby contributing to improved diagnostic reliability.
(3) Post-hoc interpretability analyses are employed to evaluate the trustworthiness of learned representations. Gradient-based sensitivity visualization highlights influential spectral regions, while dictionary atoms uncover multiscale structures. Furthermore, signal reconstructions using top-N sparse coefficients γ pinpoint critical time frequency features driving model decisions, providing physically meaningful insights for fault diagnosis.

The remainder of this paper is organized as follows: Section 2 establishes the theoretical foundation, focusing on signal decomposition and convolutional sparse coding. Section 3 introduces the proposed IMAG-Net, detailing the sparse model solution, the algorithm unrolling architecture, and the interpretability analysis. Section 4 presents comprehensive experimental verifications on both simulation data and the XJTU-SY dataset. Finally, Section 5 summarizes the conclusions and outlines future research directions.

2. Relevant theories and related work

2.1. Signal characterization and decomposition

Mechanical faults, such as those in bearings, are often characterized by periodic impulsive responses, which give rise to fault-related transients in mechanical vibration signals. Prior studies have shown that these fault-related transients exhibit frequency-domain sparsity, typically appearing as equidistant frequency components near resonance frequencies [41]. For such vibration signals, multiscale decomposition can be achieved using various techniques, including Empirical Mode Decomposition (EMD) and Wavelet Transform (WT). These methods facilitate adaptive feature extraction by isolating signal components with distinct frequency characteristics. As shown in Fig. 1, the fault-induced vibration signal exhibits rotational frequency components, noise, and the distinct modulation behavior, which manifests as resonance bands centered around fault characteristic frequencies and their harmonics. These components are crucial for accurate fault classification, as they reflect the periodic impact nature of the bearing defect.

In this study, adaptive one-dimensional learnable Gabor filters are adopted as the signal decomposition component [42]. The Gabor filter is mathematically defined as the modulation of a Gaussian envelope by a cosine carrier:

$$g(t) = e^{-\frac{(t-\mu)^2}{2\sigma^2}} \cos(2\pi f t), \tag{1}$$

where $e^{-\frac{(t-\mu)^2}{2\sigma^2}}$ represents the Gaussian window function with standard deviation σ and center μ, and $\cos(2\pi f t)$ denotes a cosine function with center frequency f. As shown in Fig. 2, different values of σ determine the temporal compactness of the Gabor convolution kernel, thereby influencing its time–frequency resolution. According to the Heisenberg uncertainty principle, the product of the temporal and spectral spreads cannot be arbitrarily small; thus, improving temporal localization inevitably reduces frequency resolution, and vice versa. A smaller σ produces a temporally concentrated kernel with a wider frequency bandwidth, which may cause spectral overlap or redundancy, whereas a larger σ yields a temporally dispersed kernel with finer frequency selectivity,

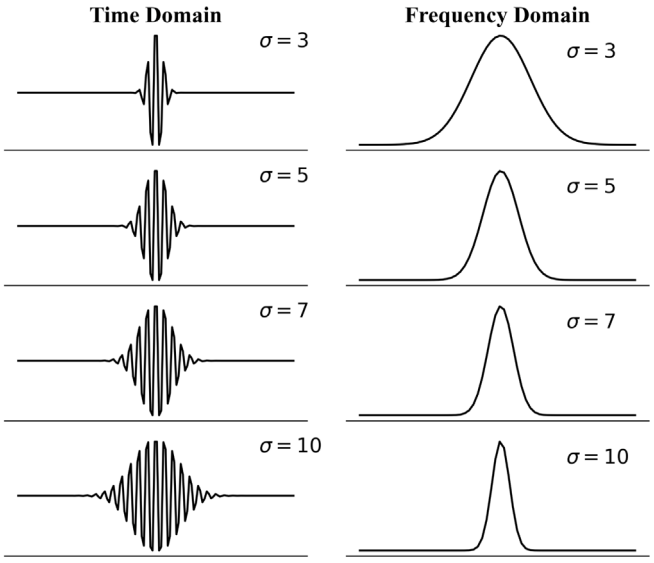


Fig. 2. Time and frequency domain performance with different standard deviation σ.

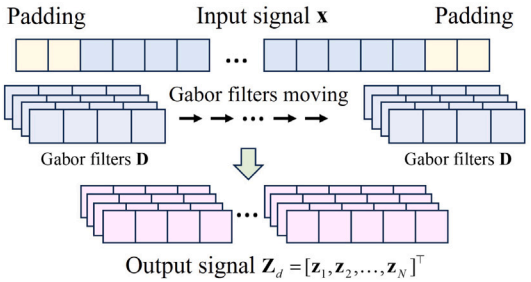


Fig. 3. Signal decomposition using learnable Gabor filters.

emphasizing detailed spectral structures. The Gabor filter therefore embodies a fundamental trade-off between time and frequency localization. To extract informative frequency components from the raw vibration signal, a bank of one-dimensional learnable convolutional filters is employed. Let $\mathbf{D}_{L\times N} = \{\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_N\}$ denote the set of learnable filters, where each filter $\mathbf{d}_i \in \mathbb{R}^L$ is designed to capture a specific frequency band and is initialized with a Gabor-shaped profile. In this method, the convolutional filters are implemented as learnable Gabor filters with trainable center frequencies and adaptive bandwidths, enabling the network to automatically adjust its frequency selectivity during training. This design allows each filter $\mathbf{d}_i$ $(i = 1, 2, \ldots, N)$ to specialize in a distinct frequency range, forming a data-driven multiscale decomposition that adaptively emphasizes discriminative frequency bands relevant to fault characteristics.

The filters are initialized in ascending order of their center frequencies, providing an ordered multiscale decomposition from low to high frequency components. Rather than remaining fixed, the center frequencies are optimized during training, enabling the filter bank to adaptively adjust its frequency responses and capture task-relevant characteristics of vibration signals. As illustrated in Fig. 3, given an input signal $\mathbf{x} \in \mathbb{R}^T$, a multiscale representation is obtained by convolving $\mathbf{x}$ with the learnable Gabor filter bank:

$$\mathbf{Z} = \mathbf{D} * \mathbf{x}, \tag{2}$$

where $*$ denotes the convolution operator and $\mathbf{D} = \{\mathbf{d}_i\}_{i=1}^N$. Symmetric zero-padding with size $p = \lfloor L/2 \rfloor$ is applied to preserve the temporal resolution, such that the output length matches that of the input signal. Specifically, the response of the ith Gabor filter at time index t is given by

$$z_i(t) = \sum_{\ell=1}^{L} d_i(\ell)\, \tilde{x}(t - \ell + p + 1), \qquad i = 1, \ldots, N, \tag{3}$$

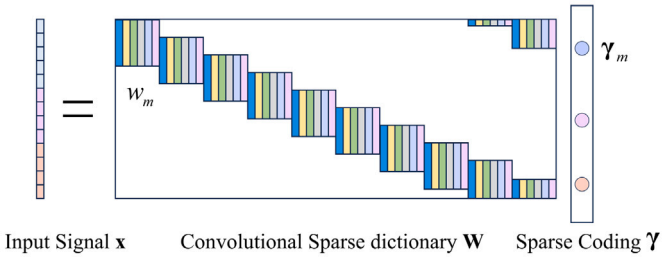


Fig. 4. Illustration of convolutional sparse coding.

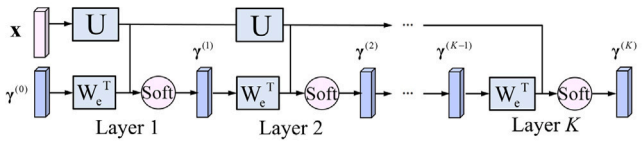


Fig. 5. The architecture of L-ISTA.

where $\mathbf{d}_i \in \mathbb{R}^L$ denotes the impulse response of the ith filter, and $\tilde{\mathbf{x}} \in \mathbb{R}^{(T+2p)}$ represents the symmetrically padded input signal. By stacking the responses of all filters along the frequency dimension, the resulting multiscale representation is expressed as

$$\mathbf{Z} = [\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_N]^\top, \tag{4}$$

where $\mathbf{z}_i$ denotes the frequency-localized response associated with the ith Gabor filter. This multiscale representation serves as the decomposed signal and is subsequently fed into the sparse coding and classification modules for interpretable fault feature extraction.

2.2. Convolutional Sparse Coding

Convolutional Sparse Coding (CSC) is a structured signal representation framework that models an observed signal as a superposition of a small number of shift-invariant convolutional atoms activated at sparse locations [43]. In this setting, sparsity implies that most coefficients remain zero while only a few take nonzero values, enabling the signal to be described by a compact set of informative components. Formally, given an input signal $\mathbf{x}$, CSC seeks a sparse decomposition over a convolutional dictionary $\{w_m\}_{m=1}^M$ by solving [44]

$$\min_{\{\gamma_m\}} \frac{1}{2} \left\| \mathbf{x} - \sum_{m=1}^{M} w_m * \boldsymbol{\gamma}_m \right\|_2^2 + \lambda \sum_{m=1}^{M} \|\boldsymbol{\gamma}_m|_1, \tag{5}$$

where w_m denotes the mth convolutional atom, $\boldsymbol{\gamma}_m$ is the corresponding sparse coding, and λ controls the sparsity level. Each atom generates translated responses along the signal axis, while sparse activations selectively encode salient local structures, as illustrated in Fig. 4. For computational convenience, the convolutional synthesis process can be equivalently expressed in a linear operator form. Specifically, all convolutional atoms and their shift-invariant structures are encapsulated in a structured convolutional sparse dictionary (CSD) $\mathbf{W}$ such that

$$\sum_{m=1}^{M} w_m * \gamma_m = \mathbf{W}\boldsymbol{\gamma}, \tag{6}$$

where $\boldsymbol{\gamma}$ denotes the global sparse representation obtained by concatenating all atom-wise sparse coefficients. To solve this optimization efficiently, we adopt the Learned Iterative Shrinkage–Thresholding Algorithm (L-ISTA) [45], which unfolds the classical ISTA into a trainable neural network. Unlike conventional ISTA, L-ISTA introduces learnable matrices to enhance adaptability while preserving interpretability and sparsity. The iterative update rule at the kth step is given by:

$$\begin{aligned} \boldsymbol{\gamma}^{(k+1)} &= \mathrm{Soft}\left(\boldsymbol{\gamma}^{(k)} + \eta \mathbf{W}^\top (\mathbf{x} - \mathbf{W}\boldsymbol{\gamma}^{(k)}),\ \lambda\eta\right) \\ &= \mathrm{Soft}\left(\eta \mathbf{W}^\top \mathbf{x} + \left(\mathbf{I} - \eta \mathbf{W}^\top \mathbf{W}\right) \boldsymbol{\gamma}^{(k)},\ \lambda\eta\right) \\ &= \mathrm{Soft}\left(\mathbf{U}\mathbf{x} + \mathbf{W}_e^\top \boldsymbol{\gamma}^{(k)},\ \theta\right), \end{aligned} \tag{7}$$

where $\mathbf{U} = \eta \mathbf{W}^\top$, $\mathbf{W}_e^\top = \mathbf{I} - \eta \mathbf{W}^\top \mathbf{W}, \theta = \lambda\eta$. Here $\mathbf{U}$ and $\mathbf{W}_e$ are treated as end-to-end optimized learnable parameters during training. This parameterization enables the network to adaptively learn the encoding transformation and the iterative interaction between reconstruction and sparsity.

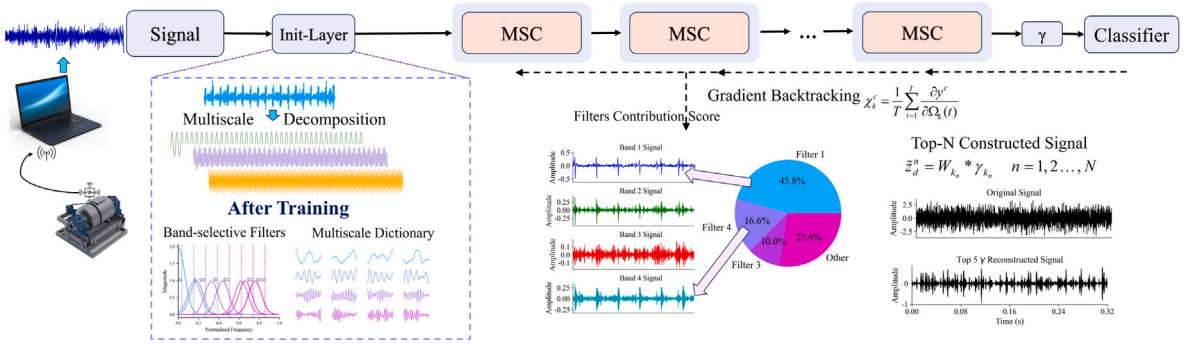


Fig. 6. The architecture of the unrolling network.

The soft-thresholding operator is applied element-wise and defined as

$$\mathrm{Soft}(a, \theta) = \mathrm{sign}(a) \cdot \max(|a| - \theta, 0),$$

where θ controls the degree of sparsity imposed on the latent representation. By unfolding ISTA into a learnable iterative network, as shown in Fig. 5, L-ISTA enables the optimization process to be jointly tuned with data through trainable parameters. The introduction of trainable parameters enables faster convergence, improved reconstruction quality, and more effective fault feature extraction [46,47].

3. Model architecture and optimization

This section presents the complete formulation and implementation of the proposed IMAG-Net. We begin by defining the problem and describing the overall network architecture, where the input signal undergoes multiscale decomposition via a learnable Gabor filter bank, followed by convolutional sparse representation and interpretable modeling. Subsequently, we introduce the optimization strategy based on model-driven sparse coding, where the convolutional sparse coding process is formulated as a constrained problem and solved via iterative methods. Finally, we transform this iterative solution into a structured neural network through algorithm unrolling, enabling layer-wise transparency and post-hoc interpretability.

3.1. Overall architecture of IMAG-Net

The proposed IMAG-Net integrates model-driven sparse representation with deep neural network architectures for vibration-based fault diagnosis. The overall framework is illustrated in Fig. 6. The network performs multiscale signal analysis, structured feature learning, fault classification, and post-hoc interpretability analysis within a unified modeling framework. Given an input vibration signal, IMAG-Net first applies a learnable Gabor filter bank to decompose the signal into frequency-selective components at multiple scales. These components are then processed by a multiscale sparse coding module implemented via an unrolled iterative structure, which progressively refines sparse and structured feature representations. The learned features are subsequently used for fault classification, while the intermediate representations enable post-hoc analysis in both the frequency and component domains. In contrast to conventional deep convolutional networks that rely purely on data-driven feature extraction, the proposed architecture is derived from a principled sparse representation formulation. Each stage of IMAG-Net corresponds to a specific operation in an iterative optimization procedure, allowing the network behavior to be analyzed and interpreted through the underlying signal processing model.

3.2. Multiscale sparse modeling formulation

To achieve interpretable and sparse multiscale representations of vibration signals, the proposed method formulates the following optimization problem:

$$\min_{\mathbf{Z},\boldsymbol{\gamma}} \frac{1}{2}\|\mathbf{D} * \mathbf{x} - \mathbf{Z}\|_2^2 + \frac{\lambda_1}{2}\|\mathbf{Z} - \mathbf{W}\boldsymbol{\gamma}\|_2^2 + \lambda_2\|\mathbf{Z}\|_1 + \lambda_3\|\boldsymbol{\gamma}\|_1, \tag{8}$$

where $\mathbf{x} \in \mathbb{R}^T$ denotes the input vibration signal. The first term enforces fidelity between the input signal $\mathbf{x}$ and its multiscale representation $\mathbf{Z}$, where the learnable Gabor filter bank $\mathbf{D}$ performs one-dimensional convolutions with $\mathbf{x}$ to extract frequency-selective components. Each filter $\mathbf{d}_i \in \mathbb{R}^L$ possesses a learnable center frequency, enabling adaptive decomposition of the vibration signal into distinct frequency bands. The second term imposes reconstruction consistency by constraining the multiscale features $\mathbf{Z} \in \mathbb{R}^{N\times T}$ to be linearly reconstructed through the convolutional sparse dictionary (CSD) $\mathbf{W} \in \mathbb{R}^{N\times M}$ and its corresponding sparse coefficients $\boldsymbol{\gamma} \in \mathbb{R}^{M\times T}$. This formulation enforces structured interactions among different frequency components while preserving the

interpretability of the learned representation. The remaining ℓ_1-norm regularization terms promote sparsity in both the multiscale features $\mathbf{Z}$ and the latent coefficients $\boldsymbol{\gamma}$, with $\lambda_1, \lambda_2, \lambda_3 \geq 0$ controlling the trade-off between reconstruction accuracy and sparsity regularization. Overall, the objective in (8) defines a structured multiscale sparse modeling framework, in which the operators $\mathbf{D}$ and $\mathbf{W}$ jointly characterize the representation space. The solution of this optimization problem, as well as its correspondence to an unfolded iterative network, will be detailed in the subsequent subsections.

3.3. Model optimization and solution

To efficiently solve this optimization problem, we adopt the Alternating Direction Method of Multipliers (ADMM) framework by leveraging the separable structure of the variable $\boldsymbol{\gamma}$, which leads to the following constrained formulation [48]:

$$\begin{aligned} \min_{\mathbf{Z},\boldsymbol{\gamma},\mathbf{u}} \quad & \frac{1}{2}\|\mathbf{D} * \mathbf{x} - \mathbf{Z}\|_2^2 + \frac{\lambda_1}{2}\|\mathbf{Z} - \mathbf{W}\boldsymbol{\gamma}\|_2^2 + \lambda_2\|\mathbf{Z}\|_1 + \lambda_3\|\mathbf{u}\|_1, \\ \text{s.t.} \quad & \boldsymbol{\gamma} = \mathbf{u}. \end{aligned} \tag{9}$$

The corresponding augmented Lagrangian function can be expressed as:

$$\begin{aligned} \mathcal{L}(\mathbf{Z}, \boldsymbol{\gamma}, \mathbf{u}, \boldsymbol{\mu}) = & \frac{1}{2}\|\mathbf{D} * \mathbf{x} - \mathbf{Z}\|_2^2 + \frac{\lambda_1}{2}\|\mathbf{Z} - \mathbf{W}\boldsymbol{\gamma}\|_2^2 + \lambda_2\|\mathbf{Z}\|_1 \\ & + \lambda_3\|\mathbf{u}\|_1 + \boldsymbol{\mu}^\top(\boldsymbol{\gamma} - \mathbf{u}) + \frac{\rho}{2}\|\boldsymbol{\gamma} - \mathbf{u}\|_2^2, \end{aligned} \tag{10}$$

where $\boldsymbol{\mu}$ denotes the dual variable and $\rho > 0$ is a penalty coefficient. ADMM proceeds by iteratively updating $\mathbf{Z}$, $\boldsymbol{\gamma}$, $\mathbf{u}$, and $\boldsymbol{\mu}$. Each update step corresponds to an interpretable module within the proposed IMAG-Net, as detailed below.

3.3.1. Multi-Scale Sparse Encoding (MSSE) block for updating Z

The update of $\mathbf{Z}$ is obtained by solving the following subproblem:

$$\begin{aligned} \mathbf{Z}^{(k+1)} = \arg\min_{\mathbf{Z}} \ & \frac{1}{2}\|\mathbf{D} * \mathbf{x} - \mathbf{Z}\|_2^2 + \frac{\lambda_1}{2}\|\mathbf{Z} - \mathbf{W}\boldsymbol{\gamma}^{(k)}\|_2^2 \\ & + \lambda_2\|\mathbf{Z}\|_1, \end{aligned} \tag{11}$$

where the convolution $\mathbf{D} * \mathbf{x}$ employs symmetric padding to maintain temporal consistency. Since (11) contains a smooth quadratic term and a non-smooth ℓ_1-regularizer, the proximal gradient method is adopted. The smooth component is given by:

$$f(\mathbf{Z}) = \frac{1}{2}\|\mathbf{D} * \mathbf{x} - \mathbf{Z}\|_2^2 + \frac{\lambda_1}{2}\|\mathbf{Z} - \mathbf{W}\boldsymbol{\gamma}\|_2^2,$$

whose gradient and Lipschitz constant are derived as:

$$\nabla f(\mathbf{Z}) = (1 + \lambda_1)\mathbf{Z} - \mathbf{D} * \mathbf{x} - \lambda_1\mathbf{W}\boldsymbol{\gamma}, \quad L_{\mathbf{Z}} = 1 + \lambda_1. \tag{12}$$

With a step size $\tau = \frac{\eta}{L_{\mathbf{Z}}}$ ($\eta \in (0, 1]$), the proximal update can be expressed as:

$$\begin{aligned} \mathbf{Z}^{(k+1)} &= \mathrm{Soft}\left(\mathbf{Z}^{(k)} - \tau\nabla f_{\mathbf{Z}}(\mathbf{Z}^{(k)}),\ \tau\lambda_2\right) \\ &= \mathrm{Soft}\left(\eta^*\mathbf{Z}^{(k)} + h(\mathbf{x}, \gamma^{(k)}),\ \lambda^*\right) \end{aligned} \tag{13}$$

where the compact forms are defined as

$$\eta^* = (1 - \eta), \quad h(\mathbf{x}, \boldsymbol{\gamma}^{(k)}) = \frac{\eta}{1 + \lambda_1}\left(\mathbf{D} * \mathbf{x} + \lambda_1\mathbf{W}\boldsymbol{\gamma}^{(k)}\right),$$

$$\lambda^* = \frac{\eta\lambda_2}{1 + \lambda_1}.$$

This step adaptively balances reconstruction fidelity, feature sparsity, and multiscale correlation, forming the MSSE Block of the IMAG-Net, which effectively extracts structured and interpretable representations across frequency bands.

3.3.2. Adaptive Sparse Optimization (ASO) block for updating γ

The variable $\boldsymbol{\gamma}$ is updated by solving:

$$\boldsymbol{\gamma}^{(k+1)} = \arg\min_{\boldsymbol{\gamma}} \ \frac{\lambda_1}{2}\left\|\mathbf{Z}^{(k+1)} - \mathbf{W}\boldsymbol{\gamma}\right\|_2^2 + \frac{\rho}{2}\|\boldsymbol{\gamma} - \mathbf{v}\|_2^2, \tag{14}$$

where the auxiliary variable is defined as $\mathbf{v} = \mathbf{u}^{(k)} - \frac{\boldsymbol{\mu}^{(k)}}{\rho}$

The analytical solution is given by:

$$\boldsymbol{\gamma}^{(k+1)} = \left(\lambda_1\mathbf{W}^\top\mathbf{W} + \rho\mathbf{I}\right)^{-1}\left(\lambda_1\mathbf{W}^\top\mathbf{Z}^{(k+1)} + \rho\mathbf{v}\right). \tag{15}$$

To avoid explicit inversion, gradient descent is employed for efficient approximation:

$$\nabla g(\boldsymbol{\gamma}) = \lambda_1\mathbf{W}^\top(\mathbf{W}\boldsymbol{\gamma} - \mathbf{Z}^{(k+1)}) + \rho(\boldsymbol{\gamma} - \mathbf{v}), \tag{16}$$

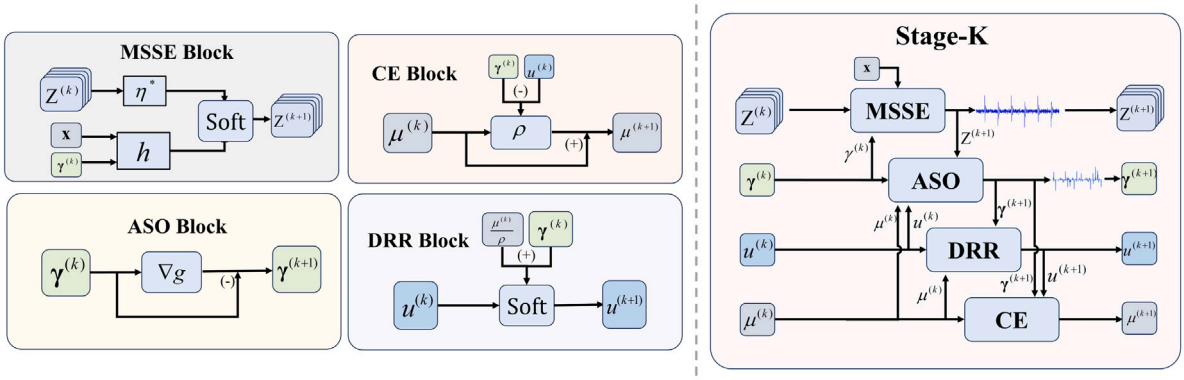


Fig. 7. The structure of one unrolled MSC block in the IMAG-Net.

with iterative update:

$$\boldsymbol{\gamma}^{(k+1)} = \boldsymbol{\gamma}^{(k)} - \eta_2 \nabla g(\boldsymbol{\gamma}^{(k)}), \tag{17}$$

where $\eta_2 > 0$ denotes the learning rate. This update step refines the adaptive sparse coefficients through learnable Gabor filters, constituting the ASO Block.

3.3.3. Dual-Residual Refinement (DRR) block for updating $\mathbf{u}$

The subproblem for $\mathbf{u}$ involves an ℓ_1-regularized quadratic minimization:

$$\mathbf{u}^{(k+1)} = \arg\min_{\mathbf{u}} \frac{\rho}{2} \left\| \mathbf{u} - \left(\boldsymbol{\gamma}^{(k+1)} + \frac{\boldsymbol{\mu}^{(k)}}{\rho} \right) \right\|_2^2 + \lambda_3 \|\mathbf{u}\|_1. \tag{18}$$

This yields a closed-form soft-thresholding update:

$$\mathbf{u}^{(k+1)} = \text{Soft}\left(\boldsymbol{\gamma}^{(k+1)} + \frac{\boldsymbol{\mu}^{(k)}}{\rho}, \frac{\lambda_3}{\rho} \right). \tag{19}$$

The DRR Block performs a soft-thresholding iteration to promote sparsity in the variable $\mathbf{u}$.

3.3.4. Consistency Enforcement (CE) block for updating $\boldsymbol{\mu}$

Finally, the dual variable is updated as:

$$\boldsymbol{\mu}^{(k+1)} = \boldsymbol{\mu}^{(k)} + \rho \left(\boldsymbol{\gamma}^{(k+1)} - \mathbf{u}^{(k+1)} \right), \tag{20}$$

which ensures the equality constraint between $\boldsymbol{\gamma}$ and $\mathbf{u}$ is progressively satisfied.

The optimization proceeds by alternately updating $\mathbf{Z}$, $\boldsymbol{\gamma}$, $\mathbf{u}$, and $\boldsymbol{\mu}$. These variable updates are unfolded as iterative modules in IMAG-Net.

3.4. Algorithm unrolling and network architecture

The IMAG-Net reformulates the ADMM-based optimization process from Section 3.3 into a trainable deep neural network through algorithm unrolling, enabling end-to-end learning with preserved interpretability. The overall architecture, depicted in Fig. 6, integrates four core components: an initialization layer for multiscale feature extraction, K Multiscale Sparse Coding (MSC) blocks for iterative sparse representation refinement, a classification module for discriminative fault diagnosis, and post-hoc analysis for interpreting fault-specific patterns. Each MSC block, illustrated in Fig. 7, encapsulates a single iteration of the ADMM updates, seamlessly connecting the optimization framework to the network structure. This total-to-part design ensures physically meaningful feature extraction and robust diagnostic performance, with details of each component provided in the following subsections.

3.4.1. Input normalization and initialization

Before feature decomposition, the input vibration signal $\mathbf{x}$ is first normalized to suppress amplitude variation and enhance the stability of the subsequent optimization process. The normalized signal is then convolved with a learnable Gabor filter bank $\mathbf{D}$, which serves as the multiscale analysis dictionary:

$$\mathbf{x}_0 = \mathbf{D} * \mathbf{x}. \tag{21}$$

Each Gabor kernel in $\mathbf{D}$ is designed to extract frequency components within specific bands, while its center frequency is adaptively learned during training, enabling the model to focus on the most informative frequency ranges.

Table 1
Network structure of the IMAG-Net.

Layer	Input shape	Output shape
Initialization	$N \times 1 \times 4096$	$N \times 8 \times 4096$
MSC Block × K	$N \times 8 \times 4096$	$N \times 64 \times 1024$
Conv-Block 1	$N \times 64 \times 1024$	$N \times 64 \times 512$
Conv-Block 2	$N \times 64 \times 512$	$N \times 128 \times 256$
Conv-Block 3	$N \times 128 \times 256$	$N \times 256 \times 128$
AdaptiveAvgPool	$N \times 256 \times 128$	$N \times 256 \times 7$
Fully connected layer	$N \times 1792$	$N \times$ Classes

3.4.2. Multiscale sparse coding via algorithm unrolling

Building on the initialized components $\mathbf{X}_0$, the unrolled network employs K MSC blocks, each corresponding to a single ADMM iteration, as depicted in Fig. 7. The right side of Fig. 7 illustrates the overall Stage K architecture, where the decomposed signal $\mathbf{Z}$ exhibits sparsity and is processed through sparse coding to generate the representation $\boldsymbol{\gamma}$. The left side details the iterative unfolding of each MSC block, which updates via four coordinated blocks. These blocks collectively execute the ADMM updates, expressed as:

$$\boldsymbol{\Theta}^{(k+1)} = \mathrm{MSC}\left(\boldsymbol{\Theta}^{(k)}\right), \tag{22}$$

where $\boldsymbol{\Theta}^{(k)} = \{\boldsymbol{\gamma}^{(k)}, \mathbf{z}^{(k)}, \mathbf{u}^{(k)}, \boldsymbol{\mu}^{(k)}\}$ denotes the set of all optimization variables at the kth iteration, and MSC$(\cdot)$ represents one unrolled step of the ADMM-based multiscale sparse coding process. Through K successive MSC blocks, the IMAG-Net refines the structured sparse representation $\boldsymbol{\gamma}$, capturing multiscale and frequency-localized fault patterns while preserving ties to the optimization framework. The final sparse maps $\boldsymbol{\gamma}$ encode interpretable time-frequency characteristics of vibration signals, which are processed by the classification module for discriminative fault diagnosis.

3.4.3. Classification module

After the multiscale sparse feature representation $\boldsymbol{\gamma}$ is obtained from the unrolled optimization process, it is further refined for classification. Although directly feeding $\boldsymbol{\gamma}$ into a fully connected (FC) classifier can preserve strong interpretability, such a design often suffers from limited discriminative capability due to the absence of hierarchical feature interaction. To achieve a balance between interpretability and discriminative power, the IMAG-Net employs a compact CNN to hierarchically process the sparse coefficients $\boldsymbol{\gamma}$, enabling local feature aggregation and cross-scale representation learning. The overall architecture of the IMAG-Net is summarized in Table 1. The input signal $\mathbf{x}$ is first convolved with the learnable multiscale dictionary $\mathbf{D}$ to initialize frequency-localized responses. Then, K^* MSC Blocks are unrolled to simulate the iterative optimization procedure, producing a structured sparse representation $\boldsymbol{\gamma}$. Subsequently, the learned sparse maps are processed by a stack of Conv-Blocks that progressively refine the discriminative representation through convolutional, normalization, and nonlinear activation operations. The detailed design of Conv-Blocks is introduced below, followed by adaptive pooling and classification layers. Each Conv-Block in IMAG-Net consists of two consecutive convolutional layers, each followed by batch normalization (BN) and ReLU activation. This design allows the network to capture both low- and mid-level fault patterns within localized temporal regions, while BN stabilizes training and accelerates convergence. Given the input feature $\mathbf{F}^{(l-1)}$, the operation of the lth convolutional layer can be formulated as:

$$\mathbf{F}_c^{(l)}(t) = \max_{t' \in \mathcal{R}(t)} \left(\sigma\left(\mathrm{BN}\left(\sum_{i=1}^{C_l} \mathbf{K}_{c,i}^{(l)} * \mathbf{F}_i^{(l-1)} + b_c^{(l)} \right) \right) \right), \tag{23}$$

where $\mathbf{K}_{c,i}^{(l)}$ and $b_c^{(l)}$ denote the convolution kernel and bias, respectively, $\sigma(\cdot)$ is the ReLU activation function, and $\mathcal{R}(t)$ defines the receptive region for max pooling. To obtain a compact and scale-invariant representation, the final convolutional feature map $\mathbf{F}^{(L)}$ is processed by an adaptive average pooling layer, which reduces the temporal dimension from V to V' while preserving global fault information:

$$\tilde{\mathbf{F}}_{n,c,t'} = \frac{V}{V'} \sum_{t \in \mathcal{I}(t')} \mathbf{F}_{n,c,t}^{(L)}, \tag{24}$$

where $\mathcal{I}(t')$ denotes the temporal indices within the t'th pooled segment. The pooled tensor $\tilde{\mathbf{F}}$ is flattened into a feature vector $\mathbf{f}$ and passed through a fully connected layer for final classification:

$$\mathbf{y} = \mathbf{W}^{\mathrm{fc}}\mathbf{f} + \mathbf{b}^{\mathrm{fc}}, \tag{25}$$

where $\mathbf{W}^{\mathrm{fc}}$ and $\mathbf{b}^{\mathrm{fc}}$ are learnable parameters. The class probabilities are obtained via the Softmax function:

$$p_c = \frac{\exp(y_c)}{\sum_j \exp(y_j)}, \tag{26}$$

and the model is trained using the cross-entropy loss:

$$\mathcal{L}_{\mathrm{CE}} = -\sum_c t_c \log p_c, \tag{27}$$

where t_c denotes the one-hot encoded ground-truth label. Although trained purely under the classification objective, the unrolled MSC module enforces interpretability by linking the learned sparse activations $\boldsymbol{\gamma}$ to physically meaningful dictionary atoms. The most active atoms correspond to key time–frequency patterns, revealing fault-specific spectral structures in the original signal.

3.5. Post-hoc interpretability via sensitivity-based attribution

To enhance model transparency and foster trustworthy decision-making, we adopt post-hoc attribution methods inspired by gradient-based sensitivity analysis [49]. These techniques aim to quantify the functional contributions of different latent representations to the model's final prediction. Let y^c denote the predicted score corresponding to fault class c, $A_k \in \mathbb{R}^T$ denote the activation of the kth frequency band, and $\gamma_m \in \mathbb{R}^{\frac{1}{4}T}$ represent the mth sparse coefficient sequence obtained via dictionary encoding. The sensitivity of the output with respect to each component is computed as:

$$S_k(t) = \frac{\partial y^c}{\partial A_k(t)}, \quad Q_m(t) = \frac{\partial y^c}{\partial \boldsymbol{\gamma}_m(t)}. \tag{28}$$

To evaluate the positive contribution of each activation or sparse code, we compute the squared product of the component and its corresponding positive sensitivity:

$$\mathrm{Eff}_{A_k} = \sum_{t=1}^{T} \left[A_k(t) \cdot S_k(t) \cdot \mathcal{H}(S_k(t))\right]^2, \tag{29}$$

$$\mathrm{Eff}_{\gamma_m} = \sum_{t=1}^{\frac{1}{4}T} \left[\boldsymbol{\gamma}_m(t) \cdot Q_m(t) \cdot \mathcal{H}(Q_m(t))\right]^2, \tag{30}$$

where $\mathcal{H}(\cdot)$ denotes the Heaviside step function that suppresses negative sensitivities, ensuring that only positively contributing elements are retained. To facilitate cross-component comparison, we normalize the importance scores as follows:

$$\hat{\mathrm{Eff}}_{A_k} = \frac{\mathrm{Eff}_{A_k}}{\sum_{k=1}^{K} \mathrm{Eff}_{A_k}}, \quad \hat{\mathrm{Eff}}_{\boldsymbol{\gamma}} = \frac{\mathrm{Eff}_{\boldsymbol{\gamma}_m}}{\sum_{m=1}^{M} \mathrm{Eff}_{\boldsymbol{\gamma}_m}}, \tag{31}$$

where K and M denote the total number of frequency bands and sparse coding, respectively. The resulting signal highlights key fault-related components, facilitating visual analysis of spectral patterns in the vibration data. To further interpret the sparse representation, we identify the top-N most influential atoms based on the highest normalized effects $\hat{\mathrm{Eff}}_{\gamma_m}$. Let $\mathcal{M} = \{m_1, \ldots, m_N\}$ be the indices of the selected atoms. The aggregated reconstructed signal is then obtained as:

$$\tilde{\mathbf{Z}}_r = \sum_{n=1}^{N} \mathbf{W}_{m_n} \boldsymbol{\gamma}_{m_n}, \tag{32}$$

where $\mathbf{W}_{m_n}$ is the m_nth CSD atom.

4. Experimental verification

In this section, the proposed method is evaluated on the simulation datasets and XJTU-SY bearing datasets to verify its diagnostic performance, parameter robustness, and interpretability. All experiments are conducted on a Windows 11 system with an RTX 4070Ti GPU using Python 3.11 and PyTorch 2.1.0. Comprehensive analysis are performed from three perspectives: classification accuracy, sensitivity to hyperparameters, and post-hoc interpretability based on learned frequency-band contributions and sparse reconstructions.

4.1. Description of the datasets

To validate the performance and interpretability of the proposed IMAG-Net, we conduct experiments on bearing fault diagnosis datasets: the simulation dataset and the XJTU-SY dataset. These datasets differ in operational conditions, fault types, and degradation stages, allowing for a comprehensive evaluation under diverse scenarios.

Simulation Construction: To rigorously evaluate the proposed model under controlled and reproducible conditions, a physically interpretable bearing fault signal is synthesized. The simulated vibration is formulated as a modulated sequence of impulsive responses corrupted by additive Gaussian noise:

$$x(t) = Q(t) \cdot \left(\sum_{k=0}^{K-1} A_k \, h(t - T_k) \right) + n(t), \tag{33}$$

where $T_k = kT + t_0$ denotes the kth impact instant, $T = 1/f_{\mathrm{f}}$ is the impact period determined by the characteristic fault frequency f_{f}, and t_0 is a random initial phase uniformly distributed in $[0, T/4]$. The impact amplitudes A_k follow a Gaussian distribution $A_k \sim \mathcal{N}(\mu_A, \sigma_A)$, reflecting stochastic variations in impact strength caused by contact uncertainty. The term $Q(t) = 1 + a\cos(2\pi f_r t)$ introduces rotational amplitude modulation associated with the shaft rotational frequency f_r. The additive noise component $n(t)$ is modeled as zero-mean Gaussian white noise, $n(t) \sim \mathcal{N}(0, \sigma_n^2)$, to emulate sensor noise and environmental interference.

The local impulse response is defined as

$$h(t) = \begin{cases} \exp\left(-\dfrac{e_R}{\sqrt{1 - e_R^2}}(2\pi f_l t)^2\right)\cos(2\pi f_l t), & t > 0, \\ 0, & t \le 0, \end{cases} \tag{34}$$

Table 2
Simulation parameters and characteristic frequencies of the bearing fault model.

Parameter	Value	Parameter	Value
f_l	3700 Hz	e_R	0.005
f_r	84 Hz	a	0.3
μ_A	1.0	σ_A	0.1
σ_n	[0, 1.5]	f_i	454.9 Hz
f_o	301.1 Hz	f_b	197.8 Hz
f_c	42.0 Hz		

Table 3
Fault types and characteristic frequencies of the XJTU-SY bearing dataset [50].

Bearing ID	Speed (Hz)	Load (kN)	Fault type	Fault characteristic frequency (Hz)
Bearing 1_1	35	12	Outer race	107.9
Bearing 1_2	35	12	Outer race	107.9
Bearing 1_3	35	12	Outer race	107.9
Bearing 1_4	35	12	Cage	13.5
Bearing 1_5	35	12	Inner race, Outer race	172.1, 107.9
Bearing 2_1	37.5	11	Inner race	184.4
Bearing 2_2	37.5	11	Outer race	115.6
Bearing 2_3	37.5	11	Cage	14.5
Bearing 2_4	37.5	11	Outer race	115.6
Bearing 2_5	37.5	11	Outer race	115.6
Bearing 3_1	40	10	Outer race	123.3
Bearing 3_2	40	10	Cage, Rolling element, Inner race, Outer race	15.4, 82.7, 196.7, 123.3
Bearing 3_3	40	10	Inner race	196.7
Bearing 3_4	40	10	Inner race	196.7
Bearing 3_5	40	10	Outer race	123.3

where f_l represents the structural resonance frequency, and e_R is the damping coefficient controlling the decay rate of oscillations. A single-sided formulation ($t > 0$) is adopted to emulate the unidirectional impulsive excitation induced by localized bearing defects. The characteristic fault frequencies f_o, f_i, and f_b correspond to the outer race, inner race, and ball element defects, respectively, as determined by the bearing geometry and shaft rotational speed. The complete parameter configuration is summarized in Table 2. A total of 2000 vibration signals are generated under five operating conditions: healthy, inner race fault, outer race fault, ball fault, and mixed fault (outer and ball defects). Each signal segment consists of 4096 sampling points acquired at a sampling rate of 12.8 kHz. To simulate diverse measurement conditions, the noise level σ_n is varied within the range [0, 1.5].

XJTU-SY Dataset: The XJTU-SY bearing dataset provides run-to-failure vibration signals collected from 15 rolling element bearings (LDK UER204) under accelerated degradation testing [50]. The data is sampled at 25.6 kHz at one-minute intervals under three distinct operating conditions. Table 3 summarizes the fault types, operating speeds, and corresponding fault characteristic frequencies. In this work, a 15-class classification task is formulated, where each bearing is treated as an individual class. Specifically, we utilize vibration data from the final 20 min of each bearing's operational life, representing a late-stage phase where evident fault conditions have already manifested. This setup serves as a fine-grained diagnosis task to evaluate the model's ability to discriminate subtle differences in fault characteristics across independent bearing runs. The data are split in a chronological manner, with the first 80% used for training and the remaining 20% reserved for testing. This setup ensures that the model learns from late-stage degradation patterns while being evaluated on temporally unseen failure progression, providing a realistic assessment of its diagnostic performance.

4.2. Experimental results and discussion

To validate the effectiveness and generalization capability of the proposed IMAG-Net, extensive experiments were conducted on both the simulation dataset and the real-world XJTU-SY bearing dataset. The comparative results against representative baseline models are summarized in Table 4. The chosen baselines encompass standard deep learning architectures and recently proposed interpretable fault diagnosis models (MWA-CNN, TFN-Net) [32,51] to provide a comprehensive comparison of classification performance and model efficiency. In this section, the IMAG-Net (Backbone) denotes the ablation version that removes interpretable feature extraction modules, retaining only the CNN backbone, while the IMAG-Net (fixed) employs non-learnable Gabor filters with fixed center frequencies. The comparison analyzes classification accuracy, model complexity, and interpretability to comprehensively assess the diagnostic capabilities of the physically constrained architecture.

The results across both datasets collectively illustrate the diagnostic consistency and interpretability of the IMAG-Net. As illustrated in Fig. 8, for the simulation dataset, the IMAG-Net maintains nearly perfect classification performance under all noise levels up to $\sigma_n = 0.7$, achieving above 99.7% accuracy. Notably, even under severe noise conditions such as $\sigma_n = 1.0$, the accuracy remains above 94%, and at $\sigma_n = 1.5$ it still reaches 63.0%, outperforming other deep models. This stability under interference can be

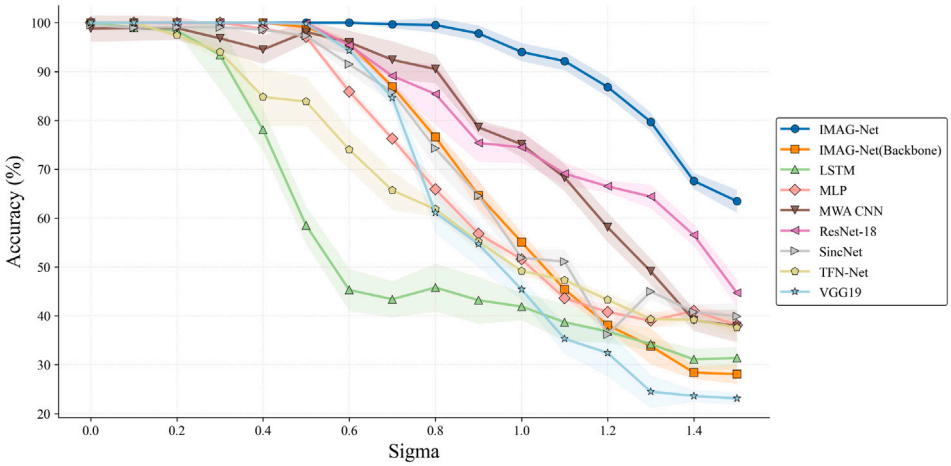


Fig. 8. Classification accuracy under different noise levels σ_n.

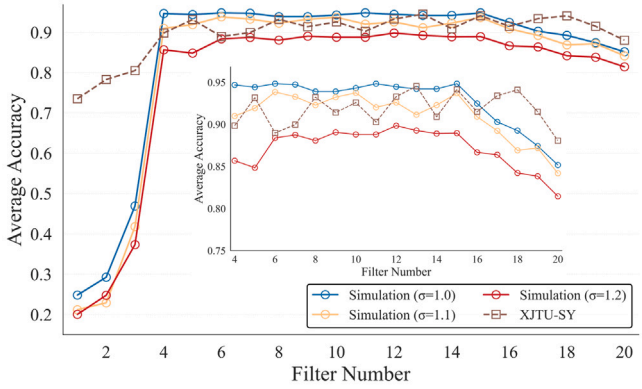


Fig. 9. Effect of the number of Gabor filters on classification accuracy.

Table 4
Comparison of model interpretability, classification accuracy on the XJTU-SY dataset, and model size.

Method	Interpretable	Accuracy (%, mean ± std)	Parameters (M)
IMAG-Net (Proposed)	✓	**92.94** ± 0.52	1.50
ResNet18		88.83 ± **0.42**	16.00
VGG19		88.05 ± 1.72	10.20
MLP		77.52 ± 0.86	2.40
MWA-CNN [51]	✓	82.96 ± 2.74	1.40
TFN-Net [32]	✓	73.45 ± 2.55	**0.80**
IMAG-Net (Backbone)		81.05 ± 2.26	1.42
IMAG-Net (fixed)	✓	89.67 ± 1.65	1.50

ascribed to the inherent sparsity prior of the unrolling architecture. By mathematically enforcing a sparse representation, the model effectively filters out incoherent background noise components while retaining the structured transients associated with faults.

In contrast, conventional architectures such as VGG-19 and ResNet-18 exhibit rapid degradation once σ_n exceeds 0.6. The IMAG-Net (Backbone) variant, which removes interpretable modules, also drops sharply at high noise. On the XJTU-SY dataset, as shown in Table 4, the IMAG-Net maintains superior performance with a compact model size of only 1.5M. While traditional CNNs rely on large-scale parameters to memorize data distributions, they are prone to redundancy. In contrast, IMAG-Net achieves stable accuracy (92.94%) with significantly fewer parameters. This efficiency stems from the frequency-adaptive nature of the Gabor kernels, which allows the network to directly parameterize the fault features using a few center frequencies rather than learning complex weights from scratch. This ensures that the model remains lightweight without compromising its ability to capture discriminative representations.

When the noise level σ_n is below 1.0, IMAG-Net maintains stable and accurate classification, while noticeable degradation appears beyond $\sigma_n = 1.3$ due to excessive noise. Therefore, the ablation experiments focus on $\sigma_n \in [1.0, 1.2]$, where the influence of key

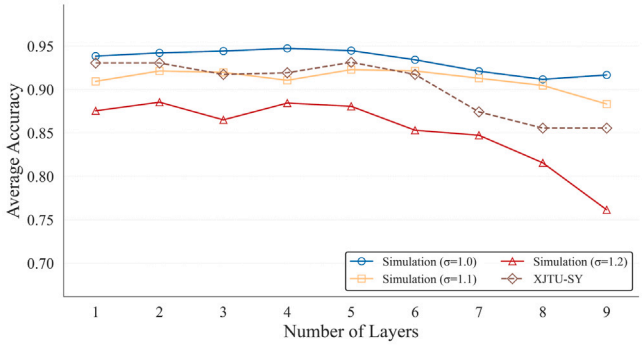


Fig. 10. Effect of the number of unrolling iterations on classification accuracy.

hyperparameters on model performance can be effectively evaluated. The analysis indicates that the number of learnable Gabor filters is crucial for balancing spectral coverage and model redundancy. As seen in Fig. 9, using fewer than four filters is insufficient to capture the spectral diversity needed for accurate classification. This is likely because bearing faults often manifest as a fundamental frequency accompanied by multiple harmonics; insufficient filters fail to capture this full harmonic series. Conversely, increasing the number of filters enhances the network's representation capability, yet exceeding sixteen filters introduces redundant subbands and overlapping responses. This leads to diminishing returns or slight degradation, suggesting that a compact set of adaptive filters is sufficient to span the essential fault-related frequency bands without introducing unnecessary computational noise.

Similarly, the unfolding depth influences convergence and feature sparsity, as illustrated in Fig. 10. Results show that a moderate number of iterations is optimal for achieving stable performance. While deeper unfolding theoretically approximates the sparse coding solution more precisely, excessive iterations may lead to performance saturation or slight degradation. This phenomenon can be attributed to the over-smoothing effect caused by repeated soft-thresholding operations, which may suppress weak but informative fault transients alongside the noise.

4.3. Model interpretablility

In the interpretability analysis, the simulated dataset with $\sigma_n = 1.1$ is employed to examine the model's behavior under high-noise conditions. As shown in Fig. 11 and Fig. 12, the evolution of the learned Gabor filter bank and the CSD is illustrated for both the simulated and XJTU-SY datasets, along with their corresponding frequency spectra, featuring CSD atoms across three representative scales. Before training, the CSD is randomly initialized, and the corresponding frequency responses appear disordered without evident multiscale organization, as evident in the scattered and overlapping spectra in (d). After training, however, the learned CSD exhibits a well-structured multiband organization, while the Gabor filters become distinctly separated and hierarchically organized, with frequency spectra showing clear peak separations and multiscale differentiation in (e) and (f). Owing to the relatively high noise level in the simulated dataset, the CSD pattern appears noisier compared to that of the XJTU-SY dataset. Nevertheless, both datasets demonstrate a clear transition from disorder to ordered structures, validating the effectiveness of the unrolling architecture in learning ordered multiscale features through gradient-based optimization.

As explicitly depicted in Fig. 12, the dynamic shift of the filter center frequencies throughout the training process further reveals interpretable adaptation behavior. For both datasets, the filters gradually tend to move toward the lower-frequency region after training and eventually stabilize in specific resonant bands, which is consistent with the findings of the frequency-principle in [22]. Specifically, as observed in Fig. 12(a) for the simulated dataset, the fifth and sixth frequency bands exhibit strong convergence, corresponding precisely to the principal resonance region around 3700 Hz, where the fault-induced oscillations are concentrated. Likewise, as shown in Fig. 12(b) for the XJTU-SY dataset, the second and third frequency bands show a similar clustering behavior, reflecting that fault-relevant information is concentrated in these lower-frequency components. These observations indicate that IMAG-Net adaptively adjusts its spectral attention in accordance with the underlying physical characteristics of vibration signals, thereby providing both interpretability and consistency with fault-related frequency phenomena.

During the envelope spectrum analysis, frequency components corresponding to one to ten times the shaft speed are excluded to reduce the influence of rotational frequencies and their harmonics. For the simulated outer-race fault signals, as illustrated in Fig. 13, the fifth and sixth frequency bands make dominant contributions to the learned representation. As shown in Fig. 14, the original vibration signal exhibits complex mixed components and a high noise level. After iterative sparse optimization through IMAG-Net, the envelope spectra of the fifth and sixth bands clearly reveal the characteristic fault frequency near the main resonance region. In the time domain, the sparse structures induced by the unrolled optimization become evident, with impulsive features corresponding to periodic fault impacts. Furthermore, the reconstructed signals based on the Top-5 γ activations effectively retain the deterministic fault transients. In contrast, less important bands such as the eighth show disordered responses with no distinct spectral peaks, reflecting the model's ability to automatically disregard uninformative frequency regions.

A comparative analysis using the ablation variant the IMAG-Net (fixed), where the Gabor filter center frequencies are non-learnable, further validates the necessity of frequency adaptivity. As depicted in Fig. 13, the fourth, fifth, and seventh bands appear

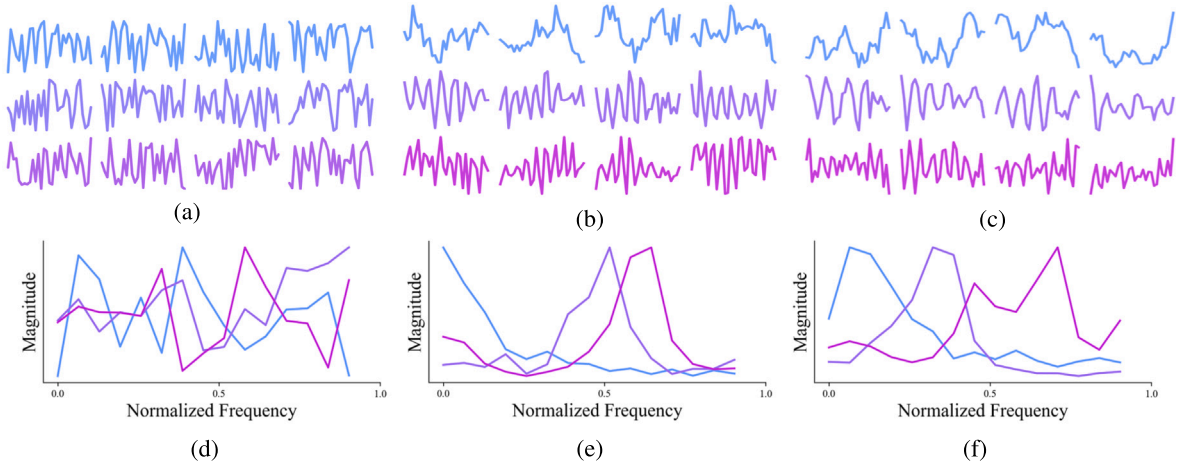


Fig. 11. Evolution of CSD during training across datasets. (a) Before training. (b) After training on the simulation dataset. (c) After training on the XJTU-SY dataset. (d) Frequency spectrum before training. (e) Frequency spectrum after training on the simulation dataset. (f) Frequency spectrum after training on the XJTU-SY dataset.

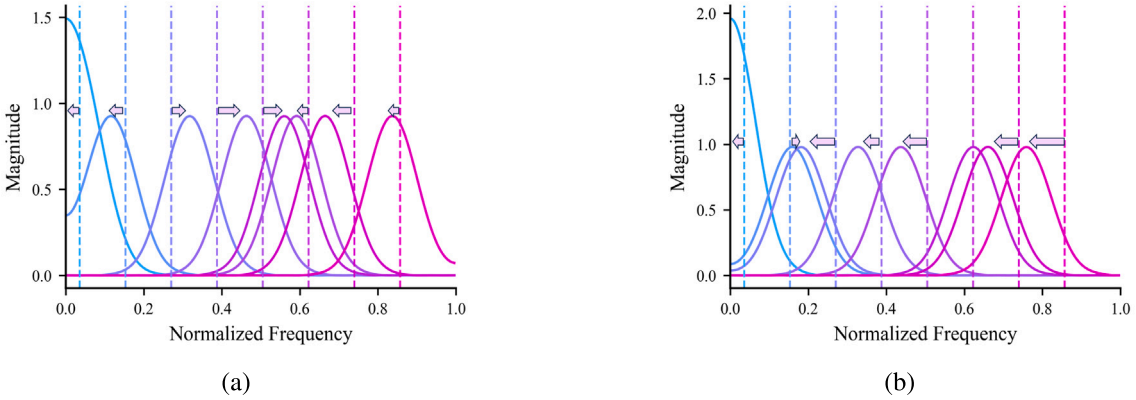


Fig. 12. Evolution of Gabor filter banks during training across datasets. (a) Simulation dataset. (b) XJTU-SY dataset.

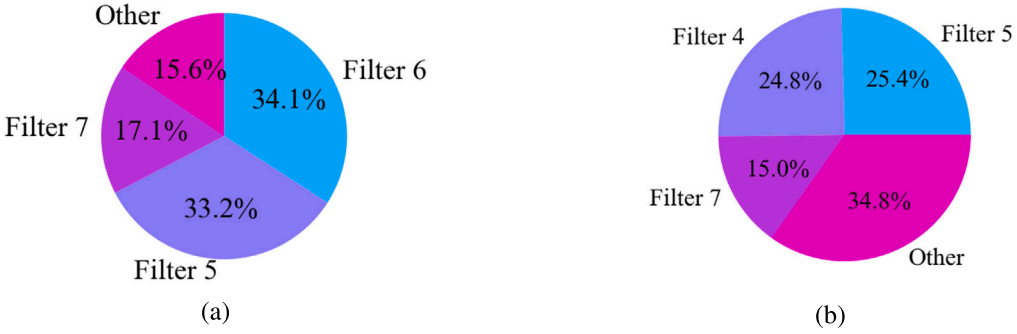


Fig. 13. Comparison of learned frequency-band contributions between the learnable and fixed versions of IMAG-Net. (a) Learnable filters in IMAG-Net. (b) IMAG-Net with fixed filters.

to have relatively high energy contributions; however, their corresponding envelope spectra in Fig. 15 are dominated by noise and lack meaningful fault-frequency components. The absence of learnable frequency adjustment prevents the model from matching the specific resonant bands of the mechanical system, fundamentally leading to reduced interpretability and lower diagnostic

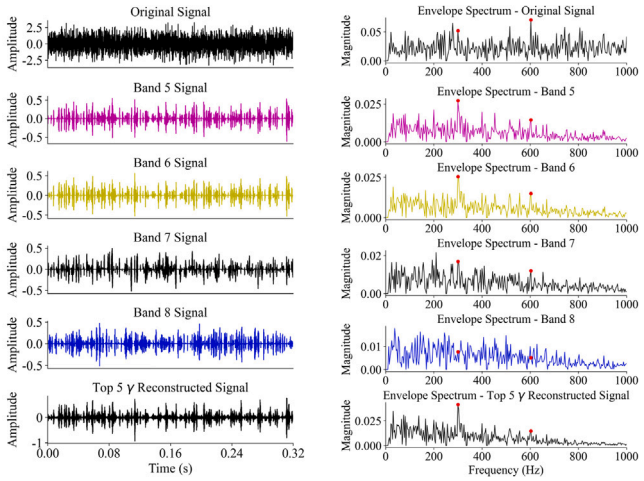


Fig. 14. Envelope spectra and time-domain signal of the key frequency bands for simulated outer-race fault signals.

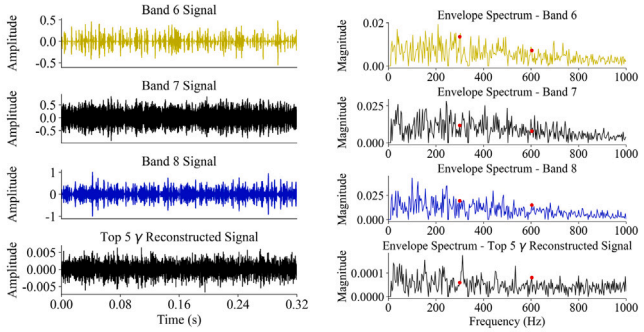


Fig. 15. Envelope spectra of the IMAG-Net (fixed) model, showing the absence of distinct fault-frequency components.

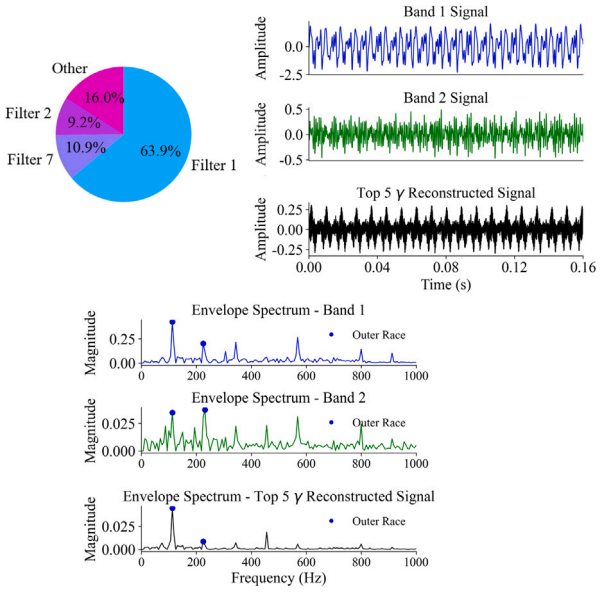


Fig. 16. Interpretability analysis for Bearing 2_4.

accuracy. These findings collectively demonstrate that the learnable Gabor filters enable the IMAG-Net to align its frequency response adaptively with the intrinsic spectral characteristics of fault signals, thereby enhancing both diagnostic precision and physical interpretability.

For the XJTU-SY dataset, the case of bearing 2_4 provides further validation. As shown in Fig. 16, the first frequency band dominates with 63.9% of the total importance. The envelope spectrum of this band exhibits strong fault-related frequency components, suggesting severe bearing degradation. Interestingly, in the time-domain signal of this first frequency band, the bearing's impulsive characteristics are clearly visible, further highlighting the presence of localized defects. These features are also preserved in the reconstructed signal, together with complementary components from the second frequency band. Such preservation of impulsive characteristics illustrates that the proposed model effectively concentrates attention on the most informative frequency bands, ensuring reliable fault identification in practical scenarios.

5. Conclusion

This study proposed the Interpretable Multiscale Algorithm Unrolling Network (IMAG-Net) to resolve the conflict between physical interpretability and accurate diagnosis under high-noise interference. By integrating learnable Gabor filters with unrolled sparse coding, the model successfully extracted physical fault features and achieved superior diagnostic performance compared to generic black-box models. Furthermore, post-hoc analyses confirmed that the decision-making process was driven by trustworthy mechanical signatures, ensuring consistent reliability even when fault features were submerged by heavy background noise.

Future work will focus on further improving the practical applicability of the proposed framework. In particular, given the evolving nature of mechanical failures, continual learning paradigms will be investigated. This may enable the model to progressively update its diagnostic knowledge for emerging fault modes while preserving previously learned patterns as much as possible, thereby enhancing long-term robustness in practical monitoring environments.

CRediT authorship contribution statement

Zhi Cao: Writing – original draft, Visualization, Validation, Methodology, Investigation. **Shibin Wang:** Writing – review & editing, Validation, Methodology, Funding acquisition. **Wei Wang:** Validation, Supervision, Methodology. **Shi-ao Wang:** Writing – review & editing, Visualization, Validation. **Haiwei Ren:** Validation, Formal analysis, Data curation. **Baoqing Ding:** Writing – review & editing, Conceptualization. **Ruqiang Yan:** Writing – review & editing, Visualization. **Xuefeng Chen:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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